Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
Department of Computer Science and Engineering
Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Persistent Event-Driven Memory Cache Module

Name of Students:

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Name of the Supervisor: Prof. Suhas Kakade

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
28/07/2026 - 03/08/2026	**Week 1 - Problem Definition and Architecture** • Finalized the problem statement: normal in-memory caches are fast but volatile and may lose acknowledged updates after a process crash. • Defined the objective of a persistent, event-driven, memory-resident key/value cache combining fast access with durable recovery. • Studied Write-Ahead Logging (WAL), crash recovery, event-driven design, LRU/LFU eviction and transaction support. • Prepared the initial modular architecture: cache engine, WAL/persistence, event bus, transaction manager, B-tree index, metrics and CLI. • Defined the critical write order: validate -> WAL first -> memory update -> publish event.	1. _____ 2. _____ 3. _____	To be filled by Supervisor	
Next Week Plan	**Planned Work** • Refine persistence/recovery design and study related systems. • Finalize component responsibilities and Phase I project plan.		To be filled by Supervisor	